Consigna Examen Tema 1: Configuración de Red con Dos Sucursales

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6to 1era Computación

Objetivo

Diseñar y configurar una red con dos sucursales interconectadas, implementando VLANs, servidores dedicados, enrutamiento estático y control de acceso mediante ACLs.

comandos usados y diagramas de red.

Configuración de Red – Sucursal A

1) Configuración de VLANs en Switch Principal

- enable
- configure terminal
- vlan 10
- name ADMINISTRACION
- exit
- vlan 20
- name USUARIOS
- exit
- vlan 30
- name WIFI
- exit
- vlan 99
- name SERVIDORES
- exit

2) Asignación de puertos de acceso a VLANs

- interface fastEthernet0/2
- switchport mode access
- switchport access vlan 10
- exit
- interface fastEthernet0/3
- switchport mode access
- switchport access vlan 20
- exit
- interface fastEthernet0/4
- switchport mode access
- switchport access vlan 30
- exit
- interface fastEthernet0/5
- switchport mode access

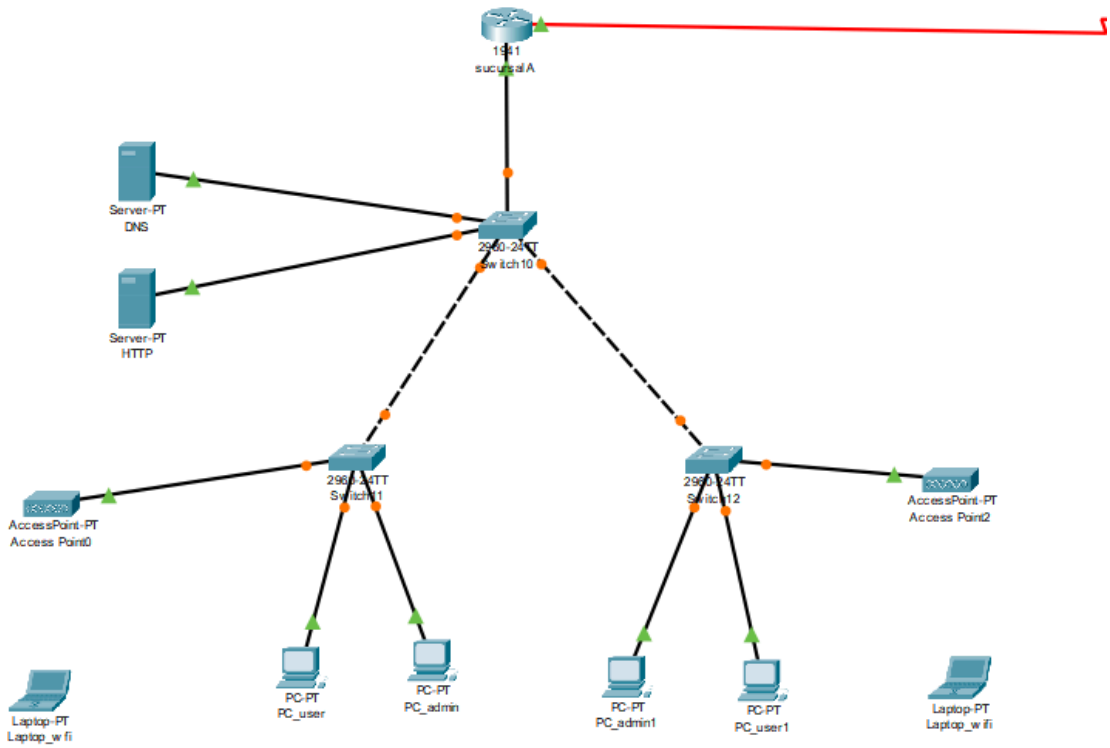
- switchport access vlan 99
- exit

3) Configuración de enlace Trunk entre Switch y Router
- interface gigabitEthernet0/1
- switchport trunk encapsulation dot1Q
- switchport mode trunk
- switchport trunk allowed vlan 10,20,30,99
- exit

4) Configuración del Router (Router-on-a-Stick)
- enable
- configure terminal
- interface gigabitEthernet0/0
- no shutdown
- exit
- interface gigabitEthernet0/0.10
- encapsulation dot1Q 10
- ip address 192.168.0.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.20
- encapsulation dot1Q 20
- ip address 192.168.1.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.30
- encapsulation dot1Q 30
- ip address 192.168.2.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.99
- encapsulation dot1Q 99
- ip address 192.168.3.1 255.255.255.0
- exit

5) Configuración del Servidor DNS y HTTP (VLAN 99)
- IP del servidor:
- IP Address: 192.168.3.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1
- DNS Server: 192.168.3.10

- DNS:
- - Services → DNS → Activar DNS Service: ON
- - Agregar registro:
- Name: empresa.local
- Address: 192.168.3.10
- HTTP:
- - Services → HTTP → Activar HTTP Service: ON
- - (Opcional) Editar index.html para personalizar la página interna



Configuración de Red – Sucursal B

1) Configuración de VLANs en Switch Principal B

- enable
- configure terminal
- vlan 10

- name ADMINISTRACION
- exit
- vlan 20
- name USUARIOS
- exit
- vlan 30
- name WIFI
- exit
- vlan 99
- name SERVIDORES
- exit

2) Asignación de puertos de acceso a VLANs
- interface fastEthernet0/2
- switchport mode access
- switchport access vlan 10
- exit
- interface fastEthernet0/3
- switchport mode access
- switchport access vlan 20
- exit
- interface fastEthernet0/4
- switchport mode access
- switchport access vlan 30
- exit
- interface fastEthernet0/5
- switchport mode access
- switchport access vlan 99
- exit

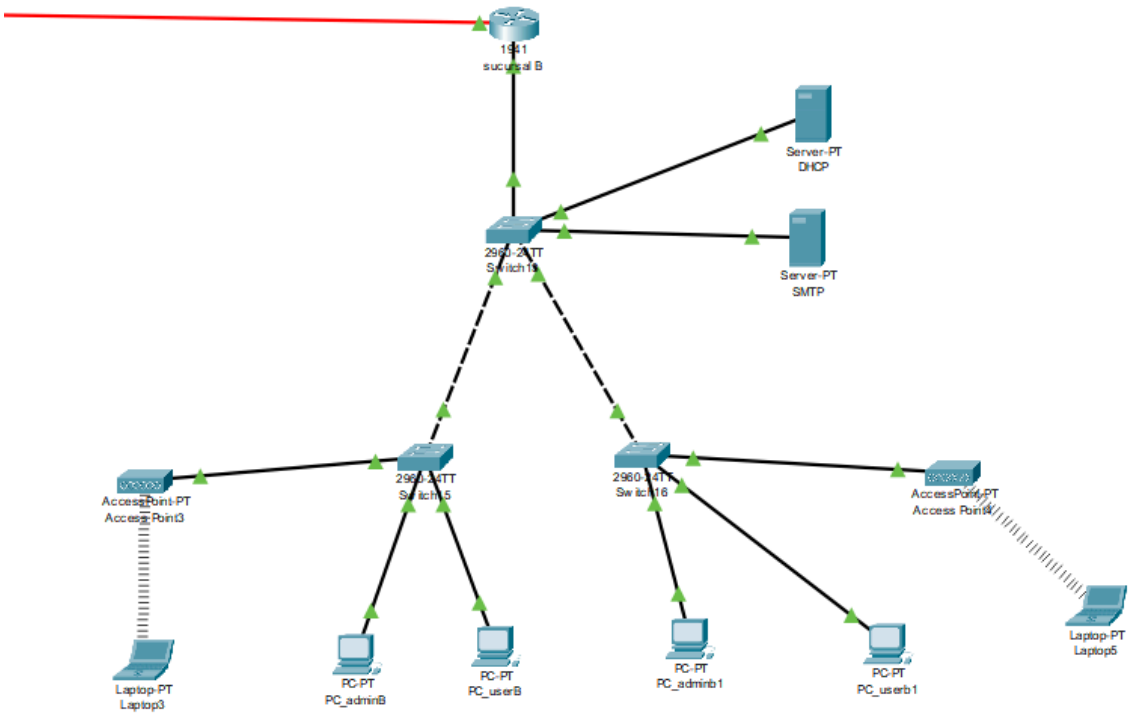
3) Configuración de enlace Trunk en Switch Principal B
- interface gigabitEthernet0/1
- switchport trunk encapsulation dot1Q
- switchport mode trunk
- switchport trunk allowed vlan 10,20,30,99
- exit

4) Configuración del Router B (Router-on-a-Stick)
- enable

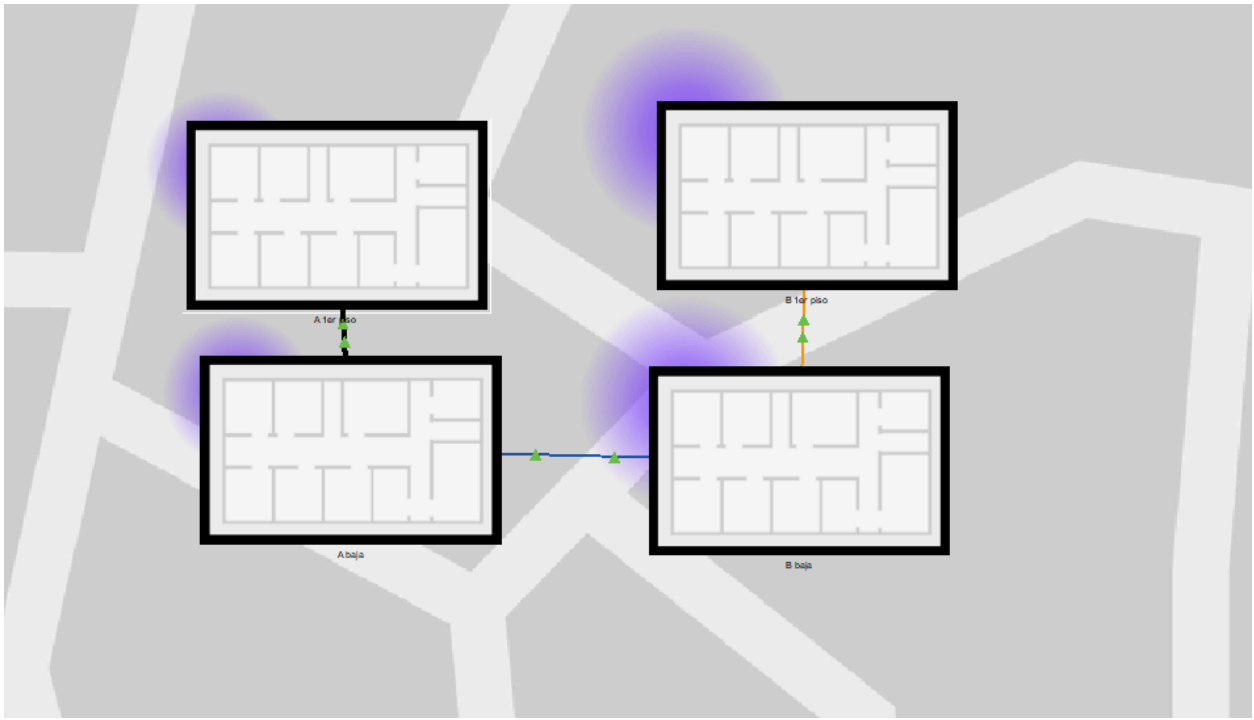
- configure terminal
- interface gigabitEthernet0/0
- no shutdown
- exit
- interface gigabitEthernet0/0.10
- encapsulation dot1Q 10
- ip address 172.19.0.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.20
- encapsulation dot1Q 20
- ip address 172.19.1.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.30
- encapsulation dot1Q 30
- ip address 172.19.2.1 255.255.255.0
- exit
- interface gigabitEthernet0/0.99
- encapsulation dot1Q 99
- ip address 172.19.3.1 255.255.255.0
- exit

5) Configuración del Servidor DHCP y SMTP (VLAN 99)

- IP del servidor:
- IP Address: 172.19.3.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 172.19.3.1
- DNS Server: 192.168.3.10 (Servidor DNS de Sucursal A)
- DHCP:
- Services → DHCP
- - Activar DHCP Service: ON
- - Crear Pools para cada VLAN
- Ejemplo para VLAN 10:
- Pool Name: ADMIN
- Default Gateway: 172.19.0.1
- DNS Server: 192.168.3.10
- Start IP: 172.19.0.50
- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 50
- SMTP:
- Services → EMAIL
- - Activar SMTP Service
- - Crear cuentas de correo para pruebas (usuario / contraseña)

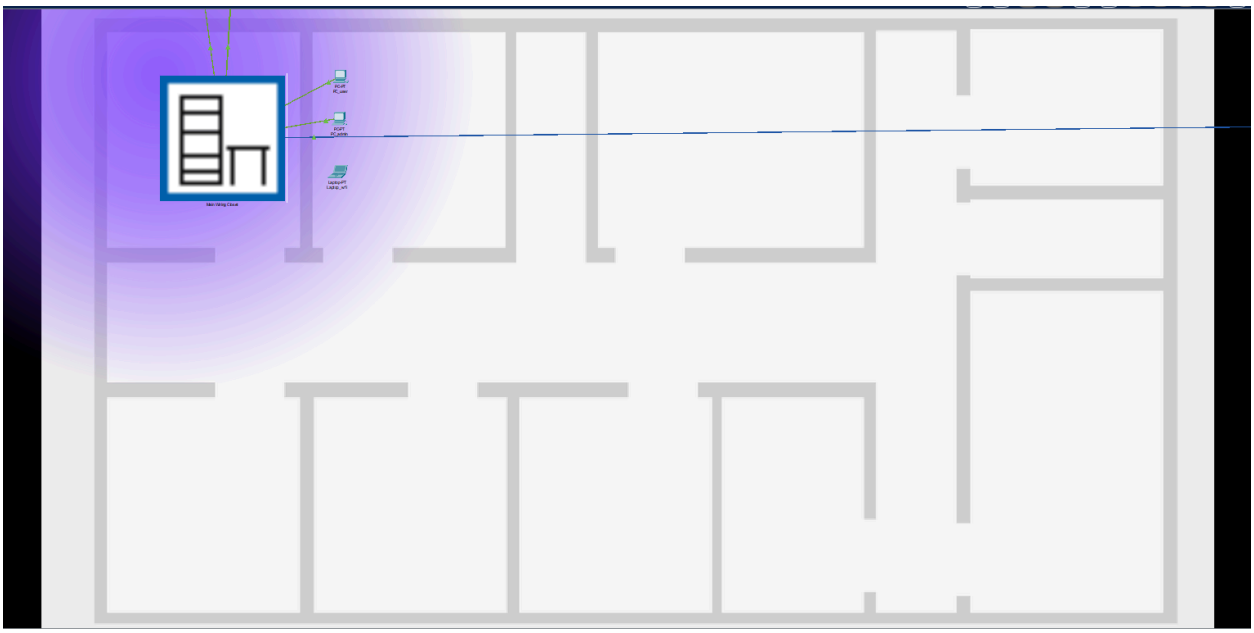


imágenes del proyecto

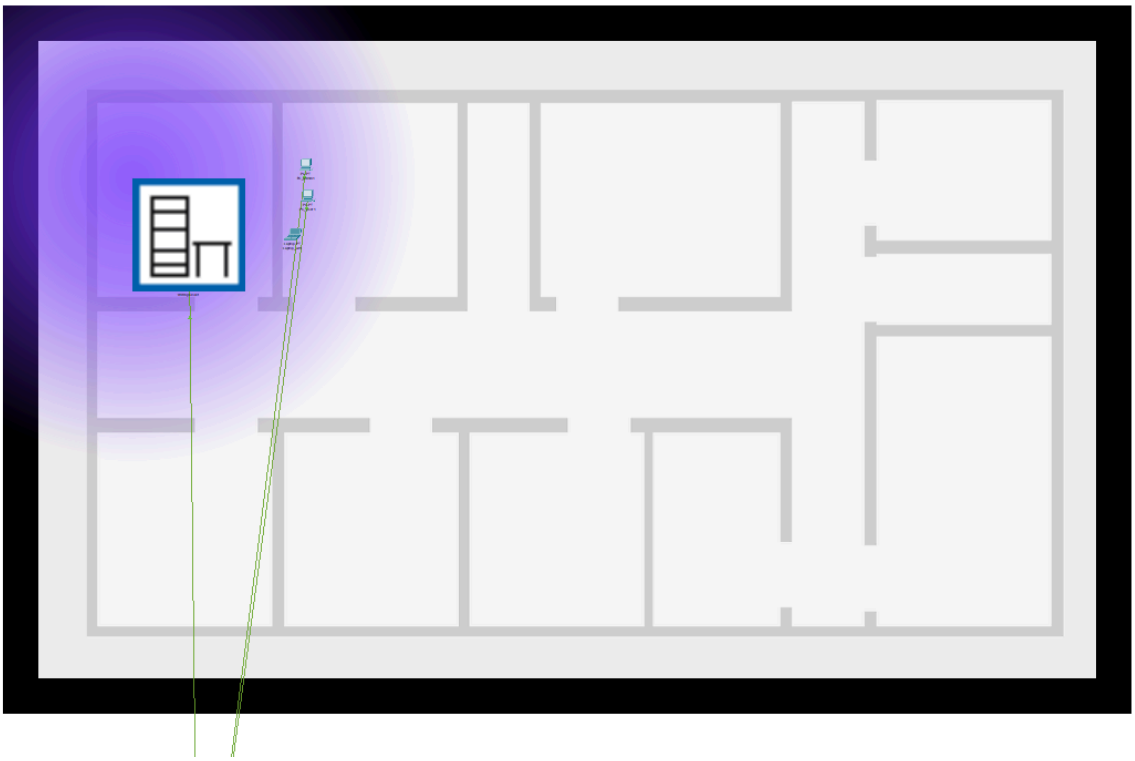


pisos de sucursal A:

piso 1

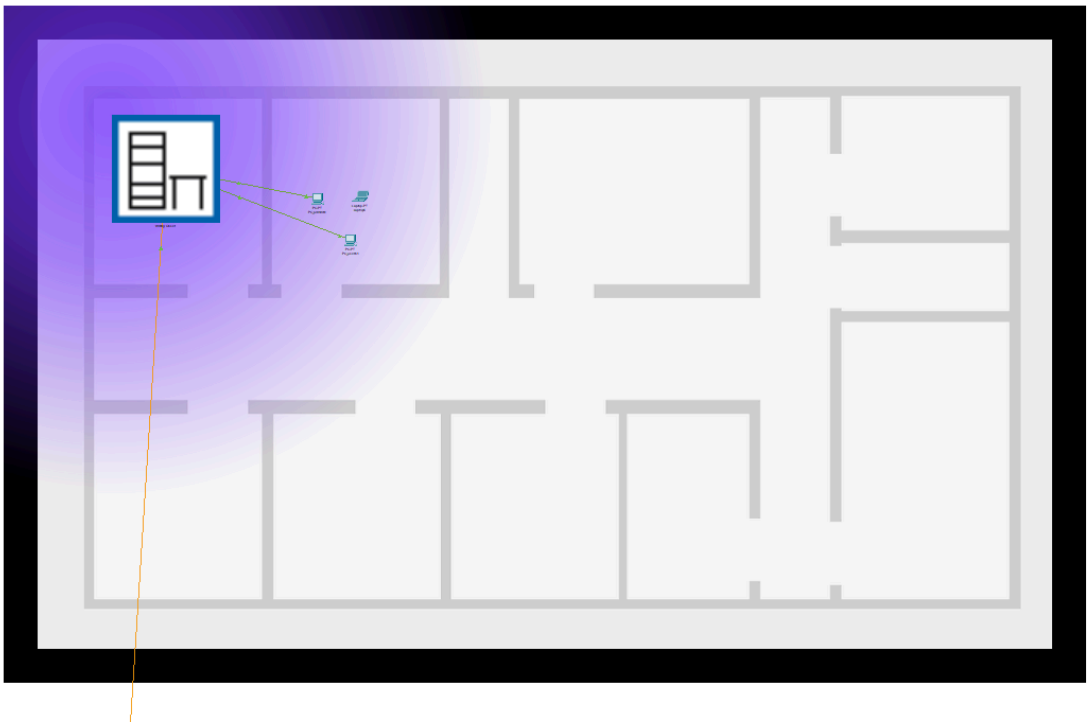


planta baja



sucursal 2

primer piso



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